Biostatistical Collaboration

Assignment 1

Question 1

Introduction:

1. Is the QoR questionnaire validated/ the standard used to measure these outcomes
2. Do you know in which journal you would like to publish? Different journals have specific requirements for the way data are collected.
3. Have the data already been cleaned and gone through some sort of quality control?
4. Were there any problems with missing data encountered?

Recruitment Stats:

1. To calculate the "recruitment rate", I would need the start and end date of the study. This will then be reported as "x patients per [time frame]"-- this can be per month/week/day etc. so I would need know that time frame. You will need the number approached and number eligible to calculate recruitment and exclusion proportions.
2. Patients excluded for language reasons might results in biased results due to the diversity of South Africa's population. We want the results of this study to be generalizable to the greater population-- is there a possibility of translating the questionnaire to several common official languages in the of the study location.

Demographics:

1. Could you clarify the definition you are using for "gender" as it can differ from the biological sex of the patient. How was this captured on RedCap?
2. Age can sometimes have extreme high or low values than can distort the mean of a sample. An alternative would be the median and inter-quartile range should that be the case.

QoR Scoring:

1. Results might be skewed. The use of mean and sd should be justified.
2. Are the QoR score categories standardized as part of the questionnaire?
3. Was the definition of "118 as patient acceptable symptom state" defined by you (the researcher) or is it a standard definition? What does this definition mean clinically.
4. The r value (known as Pearson's correlation coefficient) might not always be suitable, especially if there are outliers in the data. It will best also to calculate the Spearman correlation coefficient to better understand the relationships you are looking at.
    a. The use of plots and figures could further help identify any correlations.

   b. Correlation assesses unadjusted associations. If there is confounding (e.g., anaesthetic duration correlates with discipline), then we might consider a multivariable regression.
5. Consider regression modelling if multiple variables are involved or suspected confounders exist.

Question 2

Dear Dr Keen Researcher,

Thank you very much for your email, and for sharing the protocol and dataset. I look forward to collaborating with you on this project and helping prepare your results for publication. I hope to be helpful to you, so if there is any way I can best help you that is not covered in this email, please let me know.

If I understand you correctly, your main research goal to describe the *Quality of Recovery* of patients to better understand the satisfaction of patients after their operations, and any possible patterns related to this satisfaction when comparing different types of people and surgeries etc.

I can see your study has the potential to contribute meaningfully to the field of anaesthesia and will offer interesting insights in the sphere of post-operative outcome. I am keen to assist in making meaningful conclusions in support of your project.

To guide the way forward, I suggest the following workflow:

1. Review and clean the dataset for structure, missingness, and quality issues.
2. Confirm relevant variable definitions and coding (especially for group comparisons).
3. Conduct descriptive and inferential analyses that align with your research questions.
4. Suggest which type and the format of tables to use in your publication.
5. Discuss interpretation of the final results and to make sure these align with the journal(s) you wish to publish in.

If there are any other topics you would like me to discuss or clarify, please do not hesitate to raise these with me. To tackle my suggested workflow, I'd like to confirm a few details and raise some clarifying questions below before we proceed.

General Data Preparation

- Has the QoR-15 questionnaire been validated in the language used during administration, and was it completed without modifications?
- Has the dataset already been cleaned, and were there any known issues with data quality or missingness?
- Do you have a specific journal in mind for submission? Many journals have particular reporting requirements (e.g., STROBE), which we can tailor the output towards.

Recruitment Statistics

- To compute the recruitment rate, I will need the total study period (start and end dates), as well as the number of patients approached and deemed eligible.

- Regarding language-based exclusions: have you considered translating the questionnaire into other official languages common in your study population? This may improve generalisability and reduce bias in the selection of study participants.

Demographic Information

- Could you clarify whether the “gender” variable refers to self-identified gender or biological sex, and how it was recorded in REDCap?
- Extreme values in age or time-related variables (e.g., anaesthetic duration) can skew the mean; we may also report medians and interquartile ranges where appropriate.

QoR Scoring & Outcomes

- Are the severity classifications (excellent, good, moderate, poor) part of the standard QoR-15 instrument or based on prior literature?
- The threshold of 118 for “Patient Acceptable Symptom State” — is this a validated cut-off, or was it derived from your clinical experience?
- Given that QoR-15 is often skewed, we may evaluate the distribution first and justify the use of means and SDs accordingly.
- I am unfamiliar with the "ASA classifications". Are there any resources you would recommend for me to better prepare for subsequent meeting?
- For correlation analyses, Pearson’s r is appropriate for linear and normally distributed variables. However, we will also assess Spearman’s ρ and visualise relationships using scatterplots and smooth curves to detect non-linear trends or outliers.
- As correlation does not adjust for confounders, we may consider using multivariable regression models where overlapping influences (e.g., surgical discipline and anaesthetic duration) exist.

I appreciate the trust you’ve placed in me to support your project. My aim is to work collaboratively so that we can achieve your goals of your study whilst making sure the correct methodology is followed. Once I receive responses to the above queries and any additional recruitment data, I would be happy to schedule a 1 hour meeting at your convenience to discuss the analysis plan and timeline in more detail. It would also be helpful if your supervisor can attend this meeting.

Please don’t hesitate to share the REDCap access if you’re ready, or we can continue working from the CSV file for now. I look forward to hearing from you.

Warm regards,

Alexander van Twisk

Biostatistician

Division of Epidemiology and Biostatistics, Stellenbosch University

Question 3

1. Determine what Dr Researcher's main goals are:

- Ask Dr Researcher what the study's primary aims are in their own words.
- "What is the main message or outcome you hope to convey with this study?"
- Understand how this survey will improve patient care or inform policy decisions.
- Ask if the results will only be used for a publication or if it also forms part of their MMed dissertation.

2. To review Study Design and Recruitment, confirm the following:

- Study period: start and end dates
- Total number approached, how many were eligible, and how many were ultimately enrolled
- Reasons for exclusion, especially related to language
- Discuss implications of excluding patients on the basis of language for external validity (selection bias).
- Explore the possibility of translation of the QoR-15 tool.

3. Clarify available data and what it consists of:

- Ask if there is a codebook or similar available for variable names, types, units, and coding etc.
- Review:
    - How gender was captured (e.g., biological sex vs self-identified gender)
    - What coding was used for the surgical disciplines, ASA status, anaesthesia type, etc.
    - Ask about any known issues with data entry, cleaning, or missingness.
    - Confirm whether the current CSV is the final dataset, or if REDCap will be updated with more patients.

4. Discuss the type of analysis required:

A. For the descriptive statistics the DE asked for:

- Gender distribution
- Age (mean ± SD; also consider median/IQR if skewed)
- Surgical disciplines (frequency, percentages)
- Anaesthesia duration and time of interview (mean ± SD, range)

B. Summarising the QoR outcome

- Distribution of QoR-15 total scores: assess for normality

- Mean and SD for total score (and median/IQR if needed)
- Severity classes: frequency and percentage (ask if thresholds are standard)
- PAAS (score ≥ 118): proportion achieving this

C. Checking for associations to quantify validity of the study:

- Correlation with age, anaesthesia duration (Pearson and Spearman)
- Group comparisons:
    - Gender (t-test or Wilcoxon)
    - ASA classification (ANOVA or Kruskal-Wallis)
    - Elective vs emergency (t-test or equivalent)
    - Anaesthetic modality
    - Obstetric vs non-obstetric surgery
- Consider regression modelling if multiple variables are involved or if there are any suspected confounders.

D. Analysing different items of the research

- Identify which items had the lowest scores most frequently
- Tabulate means and SDs for each item
- Consider using figures such as boxplots to summarize results visually.
- Discuss if the patterns we see agree with clinical expectations

5. Clarify Output and Reporting Expectations

- Ask which tables/figures they would like for publication or thesis submission.